

Lab 1 - Use-Case Flow Specification

PS #36: Restaurant Table Booking & Pre-Ordering App

Use Case	UC-03 Book Table & Pre-Order
Primary Actor	Diner
Supporting Actors	Payment Gateway, Kitchen Display System
Related UCs	«include» UC-04 Make Payment, UC-07 Send Confirmation, UC-05 View Prep Timeline; «extend» UC-08 Apply Promo Code

Preconditions

- The diner is logged in to the app.
- At least one table is available for the requested date, time and party size.

Postconditions

- The table is reserved and its status is set to “Reserved” on all floor terminals.
- The pre-order ticket is queued to the Kitchen Display System with a computed prep timeline.
- Payment is authorized and a confirmation is delivered to the diner.

Main Success Scenario

1. The diner opens the app and enters date, time and party size. The system displays the interactive 2D floor plan with live table availability.
2. The diner selects an open table. The system places a temporary hold on that table.
3. The diner browses the menu and adds items to the pre-order.
4. The system computes the estimated kitchen preparation timeline and displays the target food-ready time («include» UC-05).
5. The diner confirms the reservation and pre-order. The system requests payment («include» UC-04).
6. The diner selects a payment method. The system connects to the Payment Gateway.
7. The Payment Gateway authorizes the payment.
8. The system confirms the reservation, queues the kitchen pre-order ticket, and updates the table status to “Reserved.”
9. The system sends a confirmation containing table number, date/time and pre-order summary via app notification and email/SMS («include» UC-07).
10. The use case ends successfully.

Alternate Flow — 7a: Payment Declined

- 7a1. The Payment Gateway returns a decline. The system displays a “payment failed” message and keeps the temporary table hold active.
- 7a2. The diner is prompted to try another payment method.
- 7a3. If payment fails again after 2 attempts, the system cancels the booking, releases the table back to “Available,” and notifies the diner.

Alternate Flow — 2a: Table taken during hold

- 2a1. If the held table is reserved by another diner before confirmation, the system notifies the diner and returns to the floor plan to pick another open table.